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→ Solution of a partial differential equation :-

In this we just do opposite that we have done before,
before, we formate the differential equation from a normal soln.
but, now, we will change differential equation in normal soln.

(i) Equation solvable by direct integration :

partial differential equations which can be solved by direct integration.

~~Q.1. Solve $\frac{\partial^2 z}{\partial x^2 \partial y} + 18xy^2 + 3$~~

Note → ① when we integrate w.r.t x then y is treated as constant.

② $\int 0$ w.r.t x = $f(y)$

$\left\{ \begin{array}{l} f(x) = 24 + y^2 \rightarrow \text{function of } y \\ f'(x) = 0 + 0 \end{array} \right\}$

{We will eliminate partial differential equation.}

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Q.1. Solve $\frac{\partial^3 z}{\partial x^2 \partial y} + 18xy^2 + \sin(2x-y) = 0$

=> Given;

$$\frac{\partial^3 z}{\partial x^2 \partial y} + 18xy^2 + \sin(2x-y) = 0$$

Integrate twice w.r.t. to x ($y \to$ fixed)

$$\Rightarrow \frac{\partial^2 z}{\partial x \partial y} + 18y^2 \times \frac{x^2}{2} - \frac{\cos(2x-y)}{2} = f(y)$$

$$\Rightarrow \frac{\partial z}{\partial y} + \frac{18y^2}{2} \times \frac{x^3}{3} - \frac{\sin(2x-y)}{2 \times 2} = x f(y) + g(y)$$

$$\Rightarrow \frac{\partial z}{\partial y} + 3x^3y^2 - \frac{1}{4}\sin(2x-y) = x f(y) + g(y)$$

Now, Integrate w.r.t to y ($x \to$ fixed)

$$\Rightarrow z + 3x^3 \frac{y^3}{3} - \frac{1}{4}\cos(2x-y) = x\int f(y)\,dy + \int g(y)\,dy$$

$$\Rightarrow z + x^3y^3 - \frac{1}{4}\cos(2x-y) = x \cdot u(y) + v(y)$$

$$\left[\begin{array}{l} \text{here, } u(y) = \int f(y)\,dy \\ v(y) = \int g(y)\,dy \end{array}\right]$$

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Q.2. Solve;

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{x}{y} + a.$$

$\Rightarrow$ Given;

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{x}{y} + a$$

Integrate w.r.t x ($y \to$ fixed)

$$\Rightarrow \frac{\partial z}{\partial y} = \frac{x^2}{2y} + ax + f(y)$$

Integrate w.r.t y

$$\Rightarrow z = \frac{x^2}{2}\log y + axy + \int f(y)\,dy$$

let $\psi(y)$ (say)

$$\Rightarrow z = \frac{x^2}{2}\log y + axy + \psi y$$

Q.3. Solve;

$$\frac{\partial^2 z}{\partial x \partial y} = xy$$

$\Rightarrow$ Given;

$$\frac{\partial^2 z}{\partial x \partial y} = xy$$

Integrate w.r.t x ($y \to$ fixed)

$$\Rightarrow \frac{\partial z}{\partial y} = \frac{x^2 y}{2} + f(y)$$

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Integrate w.r.t y ($x \to$ fixed)

$$\Rightarrow z = \frac{x^2y^2}{4} + \int f(y)$$

$$\Rightarrow z = \frac{x^2y^2}{4} + u(y)$$

Q.3. Solve;

$$\frac{\partial^2 u}{\partial x \partial t} = e^{-t}\cos x$$

$\Rightarrow$ Given;

$$\frac{\partial^2 u}{\partial x \partial t} = e^{-t}\cos x$$

Integrate w.r.t x ($t \to$ fixed)

$$\frac{\partial u}{\partial t} = \sin x e^{-t} + f(t)$$

Integrate w.r.t t ($x \to$ fixed)

$$u = -e^{-t}\sin x + \underbrace{\int f(t)\,dt}_{v(t)}$$

$$u = -e^{-t}\sin x + v(t)$$

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Q.4. Solve;

$$\frac{\partial^3 z}{\partial x^2 \partial y} = \cos(2x+3y)$$

⇒ Given;

$$\frac{\partial^3 z}{\partial x^2 \partial y} = \cos(2x+3y)$$

Integrate ~~twice~~ twice w.r.t x ($y \to$ fixed)

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\sin(2x+3y)}{2} + f(y)$$

$$\frac{\partial z}{\partial y} = -\frac{\cos(2x+3y)}{4} + x f(y) + g(y)$$

Integrate w.r.t y ($x \to$ fixed)

$$z = -\frac{\sin(2x+3y)}{12} + x\underbrace{\int f(y)\,dy}_{u(y)} + \underbrace{\int g(y)\,dy}_{v(y)}$$

$$z = -\frac{\sin(2x+3y)}{12} + x\,u(y) + v(y)$$

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